Qiandao Liu

Cornell University, Ithaca, NY | | | |

Education

Cornell University Aug 2023 – Dec 2026 (Expected)

B.S. in Computer Science | *Robotics Minor* | *GPA: 3.82* *Ithaca, NY*

Coursework: Independent Research (CS 4999), Robot Learning (CS 4756), Foundations of Robotics (CS 4750), Optimization Methods for Robotics (CS 5757), Deep Learning for Robotics (CS 6758).

Research Interests

My research lies at the intersection of robotics, learning, and control. I focus on enabling robots to perform useful tasks by combining novel hardware design with learning-based methods that scale with data.

Research Experience

World Labs (SceniX) Mar 2026 – Aug 2026

Robotics Research Intern *New York, NY*

Mentor: Yunzhu Li

- Led the RB-Y1 bimanual power-cord routing task as a major contributor to the real-to-sim-to-real (R2S2R) technical report.
- Served as technical lead for a Samsung Robotics proof-of-concept covering power-cord routing around a refrigerator and water-hose insertion into a refrigerator port.
- Trained ACT, Diffusion Policy, and $\pi_{0.5}$ entirely in simulation for both tasks with zero real-world training data; deployed all six policies on RB-Y1, each achieving over 90% success.
- Diffusion Policy and $\pi_{0.5}$ achieved 99.5% success in one-hour autonomous runs on both tasks.

Cornell EmPRISE Lab May 2024 – Present

Undergraduate Research Assistant *Ithaca, NY*

Advisor: Tapomayukh Bhattacharjee

- Built learning-based systems for caregiving tasks across different robots and end effectors.

Cornell Praxis Lab Aug 2025 – Dec 2025

Research Assistant (Lab Rotation) *Ithaca, NY*

Advisor: Preston Culbertson

- Worked on dexterous assembly and hand control using the LEAP Hand and WUJI Hand v1.

Publications

From Hand to Hand: A Shared Wearable Interface and Functional Benchmark for Dexterous Manipulation

Qiandao Liu*, Shengmiao Jin*, Laura Ren, Tapomayukh Bhattacharjee

*Equal contribution

Dex-CoRL Workshop 2026

Publications (continued)

Real-to-sim-to-real (R2S2R) as a scalable engine for training and evaluating robot policies
Contribution: led the RB-Y1 power-cord task.
World Labs Technical Report, 2026

RAG-Diff: Adapting Diffusion Policies to Dynamic Constraints with Retrieval-Augmented Guidance
Ruolin Ye, Nayoung Ha, Shuaixing Chen, Qiandao Liu, Gavin Chen, Shaoyang Stassen, Mark Zolotas, Jose Barreiros, Tapomayukh Bhattacharjee
Robotics: Science and Systems (RSS), 2026

Research in Preparation

Long-horizon Dressing with T-shirts using Learned Cloth Bunching and Force-limited Diffusion Policies
Pranav N. Thakkar, Qiandao Liu, Rolando Rodriguez, Keya Aggarwal, Shengmiao Jin, Tapomayukh Bhattacharjee
In Preparation

Community Service

PhyRC 2027: Physical Robotic Caregiving Challenge 2026 – Present
Organizing Team; IEEE ICRA 2027 Competition

Reviewer: Dex-CoRL 26, ICRA 27

Teaching

CS 4750: Foundations of Robotics Aug 2025 – Dec 2025
Teaching Assistant

CS 1110: Introduction to Computing Using Python Feb 2024 – May 2024
Teaching Assistant

Honors & Awards

Cornell Bowers Undergraduate Research Experience (BURE) — $7,000 research stipend 2024

Cornell RCareWorld Hackathon — 2nd Place 2024

Cornell AppDev Hackathon — Best Backend (Team) 2023

Technical Skills

Programming: Python, C++, Java, C#, MATLAB, SQL, R, Swift, HTML/CSS/JavaScript
Libraries & Frameworks: PyTorch, TensorFlow, NumPy, OpenCV, NVIDIA Warp
Simulation: NVIDIA Isaac Sim, MuJoCo, Unity
Software: ROS, ROS 2, Linux
Robot Platforms: RB-Y1, YAM, xArm6, Franka Panda, Franka Research 3, Kinova Gen3, UR5e
Dexterous Hands: LEAP Hand, WUJI Hand v1